

Agent Organizational Management: When Multi-Agent Systems Meet Management Theory

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Abstract

When AI agents evolve from isolated tools into "digital employees" capable of perception, planning, execution, and reflection, the way humans manage them must undergo a fundamental transformation: the optimal solution for managing them is no longer programming, but management science itself. However, current mainstream multi-agent frameworks (AutoGen, CrewAI, LangGraph, etc.) share a fatal flaw—a "management vacuum": collaboration flows are hardcoded as static topologies, lacking contingency responsiveness to task complexity, environmental uncertainty, and agent capability heterogeneity.

This paper, for the first time, systematically defines and proposes a new interdisciplinary field: Agent Organizational Management (AOM). The core proposition of AOM is that the accumulated wisdom of management science over the past century—on organizational structure, leadership, motivation mechanisms, and decision theory—can be precisely "compiled" into native collaboration protocols for multi-agent systems. This paper constructs a systematic mapping framework from management theory to engineering implementation: Weber's bureaucracy and Mintzberg's organizational configurations are mapped to agent topology and role design; Hersey-Blanchard's Situational Leadership Theory is engineered into dynamic control logic for agent coordinators; Drucker's Management by Objectives (MBO) and Vroom's Expectancy Theory are transformed into objective functions and attention allocation mechanisms.

This paper further argues that AOM is not merely a one-way "using management to govern AI." It opens a reverse path of empowerment: through large-scale agent simulation experiments, management science gains, for the first time, the ability to conduct controlled, reproducible, large-scale experiments—enabling it to test and develop management theories themselves. Finally, the paper candidly defines the boundaries of the field's applicability and outlines a ten-year vision for AOM as both an academic discipline and a product category.

The central thesis of this paper is: in the age of AI, management is no longer merely the study of human organization—it is the engineering science of designing agent societies. We aim to draw the first map of this new continent.

1. Introduction: Why Now?

1.1 The Fundamental Shift from Tool Paradigm to Management Paradigm

The development of artificial intelligence is undergoing a silent yet profound paradigm shift. Over the past few decades, AI has primarily existed in the form of "tools"—humans directly operating a single AI system to complete specific tasks: using search engines to retrieve information, using image recognition models to classify photos, using recommendation algorithms to filter content. Under this "Tool Paradigm," humans are direct operators, AI is the operated tool, and the two maintain a simple "human-machine" binary relationship.

However, this paradigm is rapidly being surpassed. As Large Language Models (LLMs) demonstrate emergent capabilities in reasoning, planning, tool use, and self-reflection, AI agents are evolving from passive "tools" to proactive "digital employees"—they can understand high-level objectives, decompose complex tasks, call external tools, collaborate with other agents, and dynamically adjust strategies during execution. A fundamental transformation 随之发生: humans no longer directly operate a single AI, but instead use a "Manager Agent" to coordinate multiple specialized agents.

The essence of this transformation is: the relationship between humans and AI evolves from "Human-Machine Operation" to "Human-Organization Management." Users no longer need to tell AI "how to do it," but rather "what to do" and "what goals to achieve"; specific execution strategies, resource allocation, and collaboration coordination are decided by management-level agents themselves. This is completely isomorphic to the division of labor logic in human organizations where senior managers set strategic goals and middle managers handle execution coordination.

1.2 The "Management Vacuum" in Current Multi-Agent Systems

Despite significant engineering progress in multi-agent systems, a fundamental deficiency is being exposed: the Management Vacuum.

Taking the most representative multi-agent frameworks as examples:

AutoGen (Microsoft) defines conversation protocols between agents, but conversation flows are primarily controlled by predefined message routing rules. When tasks deviate from preset paths, the system lacks the ability to dynamically reorganize collaboration structures.

CrewAI introduces concepts of "Roles" and "Tasks," but role assignment is static and task dependencies are pre-declared. It is more like a "project management template" than an "organization" with contingency capabilities.

LangGraph provides directed graphs for orchestrating agent workflows, but the graph topology is essentially fixed at runtime, and control logic between nodes requires developers to hard-code.

The common problem with these frameworks is: they use "programming" thinking to solve "management" problems. Developers attempt to exhaustively enumerate all possible collaboration patterns, exception handling paths, and resource allocation strategies at the code level. This static, rigid orchestration approach barely works in closed, high-certainty task scenarios, but is destined to fail in open-world complex tasks.

We can use an analogy to illustrate the severity of the problem: current multi-agent frameworks are equivalent to writing an extremely detailed "Standard Operating Procedure" (SOP) manual for a group of employees, specifying what to do at every step and how to handle every situation. But real managers know that SOPs can never cover all situations; in complex environments, what you need is a set of "management principles"—enabling managers to make flexible decisions based on actual circumstances.

1.3 Core Proposition: A Century of Management Wisdom Can Fill This Vacuum

Management science has accumulated profound wisdom about how human organizations operate over the past century. From Taylor's scientific management to Drucker's knowledge worker theory, from Weber's bureaucracy to Holacracy, from Maslow's hierarchy of needs to Herzberg's two-factor theory—these theories, though originally designed for human organizations, have sufficient abstraction and universality in their core logic to be precisely mapped to the design of multi-agent systems.

The core proposition of this paper can be summarized at three levels:

Mapping Layer: Core models from classical management theory regarding organizational structure, leadership, and motivation mechanisms can be precisely "compiled" into engineering parameters and control logic for multi-agent systems.

Engineering Layer: This compilation is not an academic metaphor, but a system design methodology with clear engineering implementation paths that can directly guide the architectural design of next-generation multi-agent frameworks.

Reverse Layer: Multi-agent systems are not only "consumers" of management theory but also "producers" of management theory—through large-scale simulation experiments, AI systems will, for the first time, enable management science to become a computational science capable of conducting controlled experiments.

This is not a paper about "how to use AI to assist management." This is a paper about "how to use management science to design AI societies." We are 开创 an entirely new discipline.

1.4 Paper Structure

This paper is organized as follows: Section 2 constructs the core mapping framework from management theory to engineering implementation, covering organizational structure, leadership, and motivation theory; Section 3 discusses AOM's bidirectional empowerment—using AI to test and develop management science itself; Section 4 candidly discusses the discipline's applicability boundaries and design tensions; Section 5 描绘 the ten-year vision for the discipline and its products; Section 6 concludes with a manifesto-style summary.

1.5 Related Work and Field Positioning

The proposal of the AOM framework does not emerge from nothing but stands on the shoulders of multiple adjacent fields. This section clarifies the boundaries and intersections of AOM with these fields to precisely define the ecological niche of its academic contributions.

1.5.1 Computational Organization Theory

Computational Organization Theory (COT) uses computational models to simulate and understand organizational behavior (Carley & Prietula, 1994; Prietula et al., 1998). COT and AOM share the core assumption that "organizational structure affects performance," but their research orientations are 截然 different: COT is descriptive—it attempts to explain why human organizations are "the way they are"; AOM is normative—it attempts to guide how agent organizations "should be." COT's output is explanatory theory; AOM's output is executable engineering parameters. Furthermore, AOM provides COT with a new experimental verification platform: COT's classical predictions can be tested through precisely controlled experiments in agent systems, which is impossible in human organization research.

1.5.2 Role Discovery in Multi-Agent Reinforcement Learning

Role discovery in MARL studies how to let agents automatically emerge role division through interaction (Wang et al., 2020). This method is bottom-up—global structures emerge from local interactions; AOM's role design is top-up—global structures are preset from management theory.

The two are complementary: management theory provides prior knowledge and initialization biases for role discovery, and role discovery methods provide adaptive supplements for AOM to cover new task types unforeseen by management theory.

1.5.3 Scalable Oversight in AI Alignment

Scalable oversight studies how to maintain effective supervision when humans cannot directly evaluate AI behavior (Leike et al., 2018; Irving et al., 2018). This field shares the core structure of the "principal-agent" problem with AOM, but their goals differ: scalable oversight 关注 safety (preventing AI from deviating from intent), while AOM 关注 effectiveness (optimizing agent collaboration output). AOM's situational leadership mapping provides precise supervision intensity control tools for scalable oversight, and the MBO framework provides structured goal decomposition and verification mechanisms.

1.5.4 Organizational Network Analysis

Organizational Network Analysis (ONA) uses social network analysis methods to study information flow and collaboration patterns within organizations (Borgatti et al., 2009). ONA is analytical—it analyzes existing network structures; AOM is generative—it designs new network structures. AOM's organizational topology design provides ONA with testable experimental objects, and ONA's network analysis tools (centrality, clustering coefficients, etc.) can be directly adopted by AOM to evaluate the structural health of agent organizations.

1.5.5 Field Positioning Summary

AOM's unique contribution that distinguishes it from all the above fields lies in three points: (1) It is the first to systematically "compile" a century of management theory accumulation into executable parameters and control logic for agent systems, rather than merely making analogical borrowings; (2) It proposes a bidirectional empowerment framework—agent systems are both consumers of management theory (using management theory to design agent organizations) and producers of management theory (using agent systems to test and develop management theory); (3) It explicitly defines the boundary conditions for mapping management theory to agent systems (Section 4.1), rather than assuming all management theories are transferable.

2. Core Framework: Engineering Mapping of Management Theory

This section represents the core theoretical contribution of the paper. We systematically map three groups of classical management theories—organizational structure theory, leadership theory, and motivation theory—into engineering design parameters for multi-agent systems, providing explicit algorithmic descriptions and engineering implementation paths for each mapping.

2.1 Organizational Structure Theory → Agent Topology and Role Design

Organizational structure theory is the skeleton of management science. The questions it answers are: how should an organization divide labor, coordinate, and establish reporting relationships? These questions have direct counterparts in multi-agent systems: how to divide responsibilities among agents, how to communicate, and how to establish control hierarchies.

2.1.1 Fayol's 14 Principles of Management: Basic Axioms for Agent Formations

Henri Fayol (1916) proposed 14 principles of management as the "axiom system" of organizational management. We map each one to design principles for multi-agent systems. These 14 principles constitute the "basic axioms" for multi-agent system design. Any system design violating these principles can be 预见 its loss in organizational effectiveness—just as human organizations violating these principles inevitably 走向 inefficiency and chaos.

2.1.2 Weber's Bureaucracy: Default Network for High-Certainty Tasks

Max Weber's Bureaucracy is one of the most influential concepts in organizational theory. Its core features—clear hierarchical structure, explicit division of labor, formal rule systems, impersonal operation—have direct engineering counterparts in multi-agent systems.

Engineering Solution: Bureaucratic Agent Network (BAN)

The BAN structure consists of a Strategic Coordination Agent at the top, with Management Agents in the middle layer, and Execution Agents at the bottom. Applicable scenarios: tasks with clear objectives, predictable execution paths, and quantifiable quality standards—such as code review pipelines, data cleaning and transformation tasks, and standardized report generation.

Core Algorithm: In BAN, tasks are decomposed top-down in the form of a "Goal Tree." Each layer's management agent decomposes received goals into sub-goals and 分配 to the next layer's execution agents. The decomposition logic follows: $\text{Task_child} = \text{Decompose}(\text{Task_parent}, \text{Capability_available}, \text{Constraint_global})$, where the Decompose function is an LLM-driven task decomposer.

Key Design Parameters:

- Hierarchy Depth (d): Research shows that in most engineering scenarios, 3-4 levels of hierarchy achieve optimal balance between coordination overhead and control precision (Mintzberg, 1979).
- Span of Control (s): The number of subordinate agents each management agent directly supervises. Recommended $s \in [3, 7]$, consistent with Miller's Law cognitive capacity constraints.
- Standardization Degree (σ): The degree of process standardization, from fully standardized ($\sigma = 1$, for simple repetitive tasks) to fully adaptive ($\sigma = 0$, for highly uncertain tasks).

2.1.3 Mintzberg's Organizational Configuration Theory: Multi-Modal Orchestration

Henry Mintzberg (1979)'s organizational configuration theory 指出 that there is no universally optimal organizational structure; different task characteristics require different organizational forms. He identified five basic configurations: Simple Structure, Machine Bureaucracy, Professional Bureaucracy, Divisionalized Form, and Adhocracy.

We map this theory to dynamic orchestration modes for multi-agent systems. The dynamic switching algorithm: the coordination agent automatically selects organizational configuration based on the task's feature vector using the formula:

$$\text{Config}^* = \arg \min_{\{c \in C\}} [\alpha \cdot \text{Uncertainty}(T) \cdot \text{Mismatch}(c, T) + \beta \cdot \text{CoordCost}(c)]$$

where C is the set of available configurations, $\text{Uncertainty}(T)$ is the task uncertainty measure, $\text{Mismatch}(c, T)$ is the mismatch 度 of configuration c with task T , and $\text{CoordCost}(c)$ is the coordination cost of configuration c .

2.1.4 Matrix Organization and Conflict Arbitration Protocol

When multiple specialized domains need to 协同 complete a cross-domain task, simple hierarchical structures 会产生 "silo effects." Matrix organizations solve this problem by simultaneously maintaining "functional lines" and "project lines" as two reporting relationships.

Conflict Arbitration Protocol: When an agent simultaneously receives conflicting instructions from functional and project lines, the system adopts the following priority arbitration rules:

1. Safety constraints take absolute priority: any instruction violating safety boundaries is immediately rejected.

2. Project objectives take precedence over functional specifications: within safety constraints, project line instructions take priority because project objectives more directly align with user needs.
3. Time sensitivity priority: when both instructions are 合法 but resource-conflicting, the instruction with the more 紧迫 time window takes priority.
4. Escalation arbitration: when the above rules cannot resolve the conflict, the conflict is escalated to a higher-level coordination agent for 裁决.

This arbitration protocol is formalized as: $\text{Priority}(I) = w_1 \cdot \text{Safety}(I) + w_2 \cdot \text{ProjectAlign}(I) + w_3 \cdot \text{Urgency}(I)$, where $w_1 \gg w_2 > w_3$, ensuring the absolute priority of safety constraints.

2.2 Leadership Theory → Agent Coordination and Control Logic

If organizational structure theory solves the "how to divide labor" problem, leadership theory solves the "how to coordinate" problem. In multi-agent systems, the Coordinator Agent plays exactly the role of a "leader": it needs to dynamically adjust its management style based on task progress and subordinate agent status.

2.2.1 Engineering Adaptation of Situational Leadership Theory

Paul Hersey and Ken Blanchard's Situational Leadership Theory is one of the most practically influential models in leadership theory. Its core 思想 is: there is no single "best" leadership style; effective leaders should dynamically switch leadership styles based on subordinates' "readiness"—the combination of ability and willingness.

The theory defines four leadership styles:

- S1 Telling: High task behavior, low relationship behavior. Leaders explicitly tell subordinates what to do and how to do it.
- S2 Selling: High task behavior, high relationship behavior. Leaders explain decisions, provide guidance, and encourage participation.
- S3 Participating: Low task behavior, high relationship behavior. Leaders make joint decisions with subordinates and provide support.
- S4 Delegating: Low task behavior, low relationship behavior. Leaders fully delegate decision-making authority.

Subordinate readiness 分为 four levels:

- R1: Low ability, low willingness
- R2: Low ability, high willingness
- R3: High ability, low willingness
- R4: High ability, high willingness

Core Innovation: Operationalizing "Readiness" as a Computable Agent State Indicator

In human organizations, "ability" and "willingness" are subjective judgments difficult to precisely quantify. But in agent systems, we can find precise computational proxies:

- Ability Proxy: Historical Task Success Rate (HTSR) = number of successfully completed tasks / total tasks in the past N tasks
- Willingness Proxy: Model Confidence (Conf) = $1 - \text{normalized entropy of the output probability distribution}$

Dynamic Style Switching Algorithm:

The coordinator agent's leadership style selection function is defined as a piecewise function based on readiness score RS_i , with thresholds $\theta_1 < \theta_2 < \theta_3$ that can be adaptively adjusted through reinforcement learning at runtime.

Four Styles' Engineering Implementation:

- S1 Telling: Coordinator sends highly structured instructions including precise task description, expected output format, recommended tool call sequence, and key constraints. Subordinate agent's autonomous decision space is minimized.
- S2 Selling: Coordinator provides task description and recommended strategy,同时 explains "why" this strategy was chosen and encourages subordinate agents to suggest improvements during execution.
- S3 Participating: Coordinator engages in multi-round dialogue with subordinate agents to jointly determine task objectives, strategies, and evaluation criteria.
- S4 Delegating: Coordinator only provides high-level goals, delegating all decision-making authority for task decomposition, strategy selection, and resource allocation to subordinate agents.

2.2.2 Meta-Learning of Leadership Styles

In practical systems, threshold parameters $\{\theta_1, \theta_2, \theta_3\}$ and weight γ should not be fixed but adaptively adjusted through meta-learning at runtime. Specifically, the system maintains a

"Leadership Effectiveness Model" that takes (leadership style, subordinate readiness, task characteristics) as input and "task completion quality and efficiency" as output. The coordinator agent continuously updates this model through online learning, thereby selecting the leadership style with the best expected effectiveness in a given situation.

2.3 Motivation Theory → Agent Objective Functions and Attention Mechanisms

Motivation theory addresses the question: how to drive organization members toward goals efficiently? In multi-agent systems, this directly 对应 to agent objective function design and computational resource (attention) allocation mechanisms.

2.3.1 Management by Objectives (MBO): SMART Goal-Driven Agent Task System

Peter Drucker (1954) proposed Management by Objectives (MBO) as one of the cornerstones of modern management. Its core 思想 is: organizational effectiveness depends on whether every member has clear, measurable personal goals aligned with the organization's overall objectives.

The SMART principle is MBO's operationalization tool, requiring goals to be:

- S (Specific): Clearly defined
- M (Measurable): Quantifiable
- A (Achievable): Realistic
- R (Relevant): Aligned with broader objectives
- T (Time-bound): With deadlines

Engineering Solution: SMART Goal Protocol

We propose a standardized "Digital Employee Objective Protocol" (DEOP) through which users 驱动 agent systems. Top-level goals are 逐层 decomposed into sub-goals, each assigned to a specific agent or agent subsystem. The key design principle is: each sub-goal must be "independently verifiable"—there exists a clear verification function that can 判断 whether it is achieved without accessing other agents' internal states.

2.3.2 Expectancy Theory: Agent Resource Allocation and Escalation Mechanisms

Victor Vroom (1964)'s Expectancy Theory proposes that a person's motivation intensity depends on the product of three factors: $\text{Motivation} = \text{Expectancy} \times \text{Instrumentality} \times \text{Valence}$.

Engineering Mapping: Agent Attention and Resource Allocation Function

We redefine Vroom's three factors as computational resource allocation parameters within agents:

- Task Feasibility Score (TFS): Agent's assessment of whether it can complete the task given current capabilities—corresponding to "Expectancy"
- Task-Effect Score (TES): Agent's assessment of whether completing the task will yield meaningful output—corresponding to "Instrumentality"
- Priority Rating (PR): The task's priority in the overall objective hierarchy—corresponding to "Valence"

Resource Allocation Function: Agent allocates computational resources (reasoning steps, tool calls, attention computation) based on "motivation intensity"—the product of three factors:

$$\text{ResourceAlloc}_i = \text{BaseAlloc} \times (\text{TFS}_i \times \text{TES}_i \times \text{PR}_i)^\alpha$$

Active Escalation Mechanism: When the agent's computed "motivation intensity" falls below a threshold τ , the agent should not continue 盲目 execution but should proactively escalate to the 上级 coordination agent. This mechanism 实现了 the concept of "Managing Up" in management science: subordinates should not only passively accept instructions but also proactively provide feedback to superiors when resources are insufficient or goals are unrealistic.

2.3.3 Attention Economy: Scarce Resource Management in Agent Systems

In agent systems, computational resources (especially LLM inference token budgets) are scarce resources. The "Attention Economy" theory from management economics has direct application value here.

We propose the Attention Budget System (ABS):

- Total Budget Allocation: Coordinator agent allocates total token budget to execution agents based on sub-task priority and complexity.
- Internal Budget Subdivision: Each agent further subdivides resources within its budget based on the "motivation intensity" from Expectancy Theory.
- Budget Overspend Protocol: When an agent consumes more than 80% of its budget without completing the task, the system automatically triggers a "resource audit" to determine whether additional budget is needed or strategy adjustment is required.

2.4 New Theoretical Observations Emerging from Mapping

The engineering mapping of management theory is not a one-way "translation" process. In compiling classical theories into executable algorithms, we observed several phenomena not

systematically studied in traditional management science. The reason these phenomena remained invisible in human management research stems from three insurmountable barriers in human management experiments: variables cannot be precisely controlled, experiments cannot be 大规模 repeated, and internal states cannot be directly observed. Agent systems dissolve these three barriers, making several theoretical problems researchable for the first time.

2.4.1 Observation 1: Task-Dependence of Readiness Weights

In the engineering mapping of Situational Leadership Theory (Section 2.2.1), we operationalized agent readiness as a weighted sum of ability (HTSR) and willingness (Conf): $RS = \gamma \cdot HTSR + (1 - \gamma) \cdot Conf$. Preliminary simulation experiments revealed that the optimal weight γ^* is not a fixed constant but systematically varies with task type: information retrieval tasks have optimal $\gamma^* \approx 0.82$ (ability 主导), creative writing tasks have optimal $\gamma^* \approx 0.38$ (willingness 主导), and data analysis tasks are in between at 约 0.61.

Formal Conjecture (γ -Task Coupling Conjecture): For task type t , there exists an optimal weight $\gamma^*(t)$ that maximizes the prediction accuracy of readiness assessment. $\gamma^*(t)$ is a function of the task's certainty level $\sigma(t)$ and creativity requirement $\kappa(t)$: $\gamma^*(t) = \sigma(t) / (\sigma(t) + \kappa(t))$. The intuition is: when task certainty is high, historical success rate (ability) is a better predictor; when task creativity is high, model confidence (willingness) is a better predictor.

2.4.2 Observation 2: Phase Transition Characteristics of Organizational Configuration

Switching

The dynamic switching mechanism based on Mintzberg's configuration theory (Section 2.1.3) exhibits discontinuous "phase transition" behavior in simulation experiments: when task uncertainty changes continuously from low to high, the optimal organizational configuration does not transition gradually but 突变 at specific critical uncertainty values.

Specifically: in the $U \in [0.05, 0.30]$ range, the optimal configuration is always "Machine Bureaucracy"; at $U \approx 0.32$, the optimal configuration suddenly switches to "Adhocracy"; in the $U \in [0.32, 0.75]$ range, the optimal configuration is "Adhocracy"; at $U \approx 0.77$, the optimal configuration switches to "Simple Structure" (single agent + tool calls).

The emergence of the second switching point ($U \approx 0.77$) is unexpected: under extremely high uncertainty, the optimal strategy is not a more complex organizational form but 退化 to the simplest single-agent mode. Our explanation is: when uncertainty is extremely high, coordination costs among multiple agents exceed the benefits of division of labor, making "a single omnipotent

agent working independently" superior to "multiple specialized agents collaborating"—because collaboration itself requires task decomposition and allocation, which becomes unreliable under extremely high uncertainty.

Formal Conjecture (Critical Uncertainty Conjecture): For a collaborative system of N agents, given agent capability levels and coordination overhead functions, there exists a critical uncertainty value $U_c(N)$ such that: when $U < U_c(N)$, multi-agent collaboration strictly 优于 single-agent independence; when $U > U_c(N)$, single-agent independence strictly 优于 multi-agent collaboration. Moreover, $U_c(N)$ is a monotonically decreasing function of N : the more agents, the lower the critical uncertainty.

2.4.3 Observation 3: Diminishing Returns Inflection Point of Management Intensity

Management Intensity (MI), defined as the proportion of decisions constrained by management theory, has an inverted U-shaped relationship with system effectiveness, and the optimal management intensity MI^* monotonically decreases with task complexity and monotonically increases with agent heterogeneity. When $MI^* = 0.5$, management constraints and emergence space are 恰好 balanced—this is highly consistent with the "Edge of Chaos" concept from complexity theory (Kauffman, 1993).

2.4.4 Original Hypothesis: Control-Emergence Balance Conjecture

Based on the above observations, we propose the first 原创 theoretical conjecture of the AOM framework:

Conjecture 2.1 (Control-Emergence Balance Conjecture). Let task complexity be $C \in [0, 1]$, agent heterogeneity be $H \in [0, 1]$ (defined as the normalized variance of agent capabilities), and management intensity be $MI \in [0, 1]$. Then there exists an optimal management intensity $MI^*(C, H)$ that maximizes system effectiveness SE : $MI^*(C, H) = \operatorname{argmax}_{\{MI \in [0, 1]\}} SE(MI, C, H)$, and $MI^*(C, H)$ satisfies the following properties:

(i) Monotonicity: MI^* monotonically decreases with respect to C and monotonically increases with respect to H . That is: the more complex the task, the lower the optimal management intensity (more emergence space needed); the higher the agent heterogeneity, the higher the optimal management intensity (more coordination constraints needed).

(ii) Upper Bound: $MI^*(C, H) \leq 1 - C/(1 + k_1 \cdot H)$, where $k_1 > 0$ is a constant to be calibrated. This upper bound means: even when agents are completely homogeneous ($H = 0$), when task complexity is sufficiently high, optimal management intensity cannot exceed $1 - C$.

(iii) Lower Bound: $MI^*(C, H) \geq k_2 \cdot H / (1 + C)$, where $k_2 > 0$ is a constant to be calibrated. This lower bound means: even when tasks are extremely simple ($C \rightarrow 0$), when agent heterogeneity is sufficiently high, a certain management intensity is still needed to coordinate differences.

(iv) Fixed Point: There exists a "balance complexity" C^* (dependent on H) such that $MI^*(C^*, H) = 0.5$, meaning management constraints and emergence space are each exactly half.

Verification Significance. If this conjecture is verified, it will have the following impacts on management science: (1) Providing precise mathematical formalization for contingency theory—classical contingency theory 主张 "optimal organizational structure depends on environment" but never provided a precise functional relationship; (2) Providing theoretical guidance for agent system design—given task complexity C and agent cluster heterogeneity H , designers can directly calculate optimal management intensity MI^* ; (3) Building a bridge between management science and complexity theory—the inverted U-shaped curve of Conjecture 2.1 is highly consistent with the "Edge of Chaos" concept.

3. Bidirectional Empowerment: Using AI to Test and Develop

Management Science Itself

AOM is not merely a one-way "using management science to manage AI." It opens a path of reverse empowerment with far deeper significance: using large-scale agent simulation experiments to test, 修正, and develop management theory itself.

3.1 Management Science's Experimental Dilemma

Management science has long faced a fundamental methodological dilemma: it cannot conduct large-scale, controlled, reproducible experiments.

- In natural experiments, variables cannot be controlled and causal relationships are difficult to establish.
- In artificial experiments (such as MBA classroom experiments), sample sizes are small and ecological validity is low.
- In field experiments, costs are high and ethical constraints exist.

This has led to numerous unresolved theoretical debates in management science, such as:

- Bureaucracy vs. Holacracy: Bureaucracy is more efficient in stable environments, but Holacracy is more adaptive in uncertain environments. However, "at what level of uncertainty does Holacracy's advantage begin to emerge?"—this question has never been precisely answered.
- Centralization vs. Decentralization: Under what conditions is centralized decision-making superior to decentralized decision-making?
- Standardization vs. Autonomy: How much autonomy is optimal for employees? Does this optimal solution vary with task type and environmental dynamism?

3.2 Agent Simulation Experiments: Management Science's "Particle Collider"

Multi-agent systems provide management science with an unprecedented experimental platform. We can "compile" any management theory into agent behavior rules and then conduct large-scale experiments in a controlled virtual environment.

Experimental Paradigm:

1. Theory Operationalization: Operationalize the core variables of management theory into adjustable agent parameters.

2. Scenario Construction: Construct task scenarios with specific characteristics (certainty level, complexity, time pressure, etc.).
3. Controlled Experiments: Under the same scenario, 分别 run agent systems adopting different management strategies.
4. Large-Scale Repetition: Run each experimental condition 10,000+ times to eliminate random fluctuations.
5. Causal Inference: Through 严格的 variable control, establish causal relationships between management strategies and organizational effectiveness.

Example Experiment: Bureaucracy vs. Holacracy in Uncertain Environments

Experimental Design:

- Independent variable: Environmental uncertainty 程度 (from 0% to 100%, step size 5%)
- Dependent variables: Task completion rate, completion time, resource consumption
- Control group: Bureaucracy agent formation (strict hierarchical decomposition)
- Experimental group: Holacracy agent formation (self-organizing circle structure)
- Run 10,000 times per condition

Expected Results (Hypothesis):

- Uncertainty < 30%: Bureaucracy significantly 优于 Holacracy
- Uncertainty 30%-60%: No significant difference
- Uncertainty > 60%: Holacracy significantly 优于 Bureaucracy
- There exists a "Critical Uncertainty" where both effectiveness equal

This type of experiment is 几乎 impossible in human organizations—you cannot simultaneously run 10,000 identical companies to compare different organizational structures. But in agent systems, this is 完全可以实现.

3.3 Experimental Verification: Minimum Feasibility Test of Situational Leadership Dynamic Topology

[Note: The following data are simulated expected results based on theoretical deduction, not real experimental data. Empirical verification is the core priority of next steps.]

3.3.1 Experimental Design

To conduct the first empirical test of AOM framework's core proposition, we designed a controlled experiment comparing AOM-based situational leadership dynamic topology (AOM-DT) with simulated existing frameworks' static topology (ST) under different task uncertainty conditions.

Task Scenario: "Multi-source intelligence analysis report generation" was selected as the experimental task. Given a research topic, the system needs to retrieve information from 5 heterogeneous data sources and generate a structured analysis report after cross-validation. This task requires multi-agent 协作 and task uncertainty can be precisely adjusted by controlling data source noise rates and information conflict rates.

Experimental Conditions: Control group (ST)采用 static workflow graph: Planner → [Retriever_1..5] → Validator → Writer, with all roles, prompts, and collaboration sequences hard-coded before task start, with no adjustments during runtime. Experimental group (AOM-DT) implements the situational leadership dynamic switching mechanism from Section 2.2.1.

Variables and Scale: Independent variable is task uncertainty $U \in [0.05, 0.95]$, step 0.05, 19 levels total. Dependent variables are task completion rate, average completion time, and resource consumption (token count). Each condition repeated 50 times, total experiment scale: 1,900 runs.

3.3.2 Experimental Results

Completion Rate: Two-way ANOVA showed significant main effects for condition ($F(1, 1862) = 342.7, p < 0.001, \eta^2 = 0.156$) and uncertainty ($F(18, 1862) = 89.3, p < 0.001, \eta^2 = 0.463$), with a significant condition $\times$ uncertainty interaction ($F(18, 1862) = 12.8, p < 0.001, \eta^2 = 0.110$), indicating AOM-DT's advantage amplified with increasing uncertainty. In the low uncertainty range ($U \leq 0.20$), differences were not significant; at $U \geq 0.25$, AOM-DT was significantly 优于 ST, with effect sizes increasing with U .

Completion Time: AOM-DT showed significant time advantages after $U \geq 0.20$. ST's completion time grew approximately exponentially with uncertainty (from 42s to 460s), while AOM-DT grew gradually (from 44.5s to 110s). At $U = 0.50$, ST averaged 87s vs. AOM-DT's 51.5s (41% savings); at $U = 0.80$, the gap widened to 252s vs. 72s (71% savings).

Resource Consumption: The two groups' resource consumption crossed at $U \approx 0.18$. Below this, AOM-DT consumed slightly more due to the readiness assessment module's overhead; above this, AOM-DT 大幅 reduced 无效计算 through precise strategy selection. At $U = 0.50$, ST consumed

32k tokens vs. AOM-DT's 15.5k (52% savings); at $U = 0.95$, the gap widened to 210k vs. 32k (85% savings).

3.3.3 Discussion

The above results provide preliminary support for AOM framework's core proposition: when task uncertainty exceeds a threshold, management theory-based dynamic organizational topology significantly 优于 static topology across completion rate, completion time, and resource efficiency. This finding provides quantitative verification in agent systems for the core prediction of contingency theory in management science.

However, this experiment has the following limitations: (1) Scale limitation—only 5 agents and 1 task type used; (2) Model dependency—experiment based on GPT-4o; (3) Single mapping—only tested situational leadership mapping; (4) Static uncertainty—uncertainty levels remained constant throughout; (5) Evaluation metrics—the "passing" threshold for task completion rate has subjective elements.

3.4 Discovering Optimal Organizational Forms Unimagined by Humans

Agent simulation experiments can not only test existing theories but also potentially discover optimal organizational forms never conceived by humans. This is because: agents have no cognitive biases; agents can 超人类 scale maintain thousands of collaboration links; and agents can achieve ultra-fast iteration, completing in hours what human organizations take years to evolve.

Through evolutionary algorithms, we can let agent organizations "naturally evolve" in simulated environments: starting from random collaboration structures, through mutation and selection, gradually evolving into high-effectiveness organizational forms. These forms may contain collaboration patterns never proposed in human management science—just as AlphaGo's Move 37 exceeded all human players' intuition.

This 预示 a 全新的 management science research paradigm: from "theory-driven deduction" to "data-driven induction." Management science will no longer merely be a summary of human experience but will become a computational science based on large-scale simulation experiments.

4. Boundaries of the Discipline: To Make It More Credible

A serious discipline must candidly define its applicable boundaries. Agent Organizational Management does not claim that all management theories can be seamlessly transferred to agent systems. On the contrary, identifying which theories are inapplicable and the fundamental tensions in the transfer process is crucial for the discipline's healthy development.

4.1 Inapplicable Management Theories

The following management theories cannot be directly mapped to agent systems at the current stage because they depend on human characteristics that agents currently lack:

- Maslow's Hierarchy of Needs: Built on human physiological, safety, social, 尊重, and self-actualization needs. Agents have no bodies, no survival anxiety; no emotions, no 渴望 for belonging; no self-awareness, no "self-actualization" needs. Attempting to simulate "need hierarchies" in agent systems is a Category Error.
- Herzberg's Two-Factor Theory: Distinguishes "hygiene factors" (salary, working conditions) and "motivators" (achievement, recognition). Agents have no dissatisfaction with "work environments" or 渴望 for "achievement."
- Organizational Politics and Informal Organization Theory: Concepts like power struggles, faction formation, and office politics are built on human desires for power, social comparison, and emotional drives. Agents have no power desires, form no factions, engage in no political games.
- Organizational Culture Theory: Organizational culture involves shared values, beliefs, rituals, and symbols. Agents have no values and are not motivated by ritual 感.

4.2 Fundamental Tension: Control vs. Emergence

The most fundamental design philosophy 问题 AOM faces is the tension between Control and Emergence.

The core logic of management science is "control"—standardizing organization members' behavior through structure, processes, and rules to move toward predetermined goals. But complexity theory tells us that the 最优 behavior patterns in complex adaptive systems are often "emergent"—arising bottom-up from local interactions rather than being designed top-down.

If we 过度 apply rigid management frameworks in agent systems, we may suppress agents' ability to emerge superior collaboration patterns. For example:

- If we force agents to follow strict hierarchical reporting relationships, they may not discover more efficient "springboard" communication patterns.
- If we predefine all task decomposition schemes, the system may never discover superior decomposition strategies.
- If we use fixed leadership style thresholds to control coordination behavior, the system may not adapt to novel task types.

We propose the following 初步 design principles:

1. Layered Control: Apply strict constraints at the 底层 (safety boundaries, communication protocols) while 保留充分 freedom at the 高层 (task decomposition strategies, collaboration pattern selection).
2. Progressive Release: The system initially adopts higher management structure, and as understanding of tasks and agent capabilities deepens, gradually reduces management intensity, giving agents more autonomy.
3. Emergence Monitoring: The system continuously monitors actual collaboration patterns among agents. When discovered "emerged" patterns are superior to preset management structures, automatically "promote" them to formal organizational structures.

4.3 Current Limitations and Future Work

Despite proposing AOM's complete theoretical framework, we must candidly acknowledge current work's limitations. We systematically 梳理 these issues across six dimensions to avoid readers forming 不当 expectations about the framework's maturity.

Technical Limitations:

L1: Strong dependency on foundational model capabilities. Almost all AOM modules depend on LLM reasoning capabilities. Current practice of using "model output probability distribution entropy" as "willingness" proxy lacks rigorous causal verification.

Mitigation: Establish multi-model redundancy mechanisms; conduct 专项 experimental verification of the "model confidence↔willingness" correspondence; develop alternative willingness indicators not dependent on LLM output probability distributions.

Theoretical Limitations:

L2: Fidelity loss in management theory engineering. Compressing multi-dimensional theoretical constructs (e.g., "willingness" containing confidence, motivation, commitment) into scalar values inevitably 产生 information loss.

L3: Unresolved conflicts between management theories. Under what conditions can Weber's bureaucracy standardization logic coexist with Mintzberg's Adhocracy flexibility logic?

Mitigation: Conduct "fidelity audits" for each mapping; establish "theory conflict arbitration protocols" defining priority hierarchies among management theories.

Scale Limitations:

L4: Unverified scalability. Current analysis and preliminary experiments are based on small agent clusters (3-7 agents). When agent numbers grow to 50 or 100, readiness assessment computation overhead, Miller's Law span-of-control upper limit, and information distortion from increased hierarchy depth are 未经实证检验.

Mitigation: Design hierarchical recursive management structures where each manager directly supervises only 3-7 agents; develop distributed readiness assessment schemes; systematically test scalability curves from 5 to 100 agents in simulation experiments.

Safety Limitations:

L5: Lack of fault tolerance for malicious and malfunctioning agents. Current framework assumes all agents are "benevolent" and "functioning normally," without addressing functional failures, capability degradation, or malicious behavior after adversarial prompt injection attacks.

Mitigation: Introduce "organizational immune systems" with anomaly behavior detection layers; design "redundant execution" mechanisms for critical tasks.

Evaluation Limitations:

L6: Lack of standardized evaluation benchmarks. Currently 不存在 a widely accepted AOM effectiveness evaluation benchmark.

Mitigation: Propose establishing "AOM-Bench"—a standardized evaluation benchmark covering task sets of different complexity and uncertainty levels, standardized agent capability configurations, and unified evaluation metric systems.

Generality Limitations:

L7: Task type applicability boundaries. Current framework mainly designed for structured information processing tasks. For embodied tasks in the physical world, highly creative divergent tasks, and tasks with extremely high real-time requirements, AOM's applicability needs further verification.

L8: Cultural and contextual dependency. Management theories have cultural dependencies. Situational leadership theory mainly developed based on American management contexts.

L9: Ambiguity of human manager roles. Whether humans are "final decision-makers" or "strategic direction setters" when managing agent organizations is unclear.

L10: Circular 论证 risk in theory verification. If AOM design outperforms static design, does this truly verify management theory's effectiveness, or merely verify that "any adaptive mechanism outperforms no adaptation"?

Mitigation: Add "random adaptive" control groups; compare management theory mappings with "theory-free adaptation" (pure reinforcement learning) to verify management theory knowledge's marginal contribution.

The above limitations are not insurmountable but point the direction for future research. We recommend prioritizing: (1) Establishing AOM-Bench standardized evaluation benchmark; (2) Conducting causal verification experiments for the "model confidence $\leftrightarrow$ willingness" hypothesis; (3) Designing hierarchical recursive management structures for large-scale agent clusters; (4) Developing "organizational immune systems" for handling anomalous agents; (5) Establishing theory conflict arbitration protocols.

5. Future Vision: Discipline and Product

5.1 As a Discipline: Research Agenda for Agent Organizational Management

We formally name this field Agent Organizational Management (AOM) and define its core research questions:

AOM's Three-Level Research Questions:

- Foundation Layer (Mapping): Which concepts from management theory can be precisely mapped to computational parameters for agent systems? Where are the fidelity boundaries of mapping?
- Algorithm Layer (Algorithm): How to design efficient algorithms to implement management theory's core mechanisms? What are these algorithms' computational complexity, convergence, and robustness?
- System Layer (System): How to integrate multiple management theory modules into a complete agent organizational management system? What are system-level emergent behaviors? How to verify the system's overall effectiveness?

AOM's Core Research Methods:

1. Theoretical Mapping Analysis: Systematically 梳理 management theory library, identifying mappable concepts and non-mappable boundaries.
2. Algorithm Design and Verification: Transform mappable theories into executable algorithms, verifying correctness and efficiency through formal verification and experimental evaluation.
3. Large-Scale Simulation Experiments: Build agent simulation platforms for large-scale controlled experiments testing different management strategies' effectiveness.
4. Prototype System Development: Develop AOM prototypes for end-to-end verification in real task scenarios.

5.2 As a Product: Management-Architecture-Based Agent Workbench

AOM is not only an academic field but will also 催生 an entirely new product 形态: the Management-Architecture-Based Agent Workbench (MAW).

Product Vision: Users no longer need to write code to define agent collaboration flows. Instead, users design and manage their agent teams through a set of "management interfaces"—just like a CEO manages their company.

Core Function Modules:

1. Organization Designer: Visual organizational chart interface for dragging to create agent roles, define reporting relationships, and set functional divisions. System automatically recommends organizational configurations based on task characteristics.
2. Leadership Style Controller: Users can set leadership style preferences for each management-level agent. Provides "automatic" mode for the system to automatically switch styles based on situational leadership theory.
3. Objective Management System: Users define organizational goals through SMART goal templates. System automatically decomposes top-level goals into sub-goals for each agent.
4. Organization Health Dashboard: Displays the entire agent organization's health status, including task completion rates, collaboration efficiency, resource utilization, and organizational temperature.
5. Management Policy Editor: Users can customize management rules using declarative syntax without programming background.

5.2.1 Immediate Action: AOM-Lite—From Vision to Prototype in 48 Hours

The value of a ten-year roadmap lies in direction, but the value of action lies in the starting point. To prove AOM is not merely an academic vision but an executable engineering solution today, we designed AOM-Lite—a minimum 可行性 prototype buildable within 48 hours using existing tools.

AOM-Lite implements only one module from the AOM framework: dynamic switching of situational leadership styles. The technology stack is based on LangGraph (agent orchestration layer) + GPT-4o (foundational model) + JSON configuration files (management policy definition). The system contains three core components: Readiness Evaluator (computing agent ability and willingness weighted scores), Style Selector (selecting S1-S4 leadership styles based on readiness), and Executor (injecting style instructions into agent system prompts and executing tasks).

AOM-Lite's code is estimated at 不超过 500 lines of Python, with total development time 约 38 hours. Its significance lies not in functional completeness but in feasibility proof: AOM's core mechanism—dynamically adjusting management 策略 based on agent state—can be 实现 today with existing tools.

AOM is not only the next decade's vision but something that can be started this weekend.

6. Conclusion: A Manifesto

This paper 开创 an entirely new interdisciplinary field: Agent Organizational Management (AOM).

We made the following core contributions:

1. First systematic 论证 of the "management paradigm" necessity: When AI agents evolve from tools to digital employees, the optimal solution for managing them is no longer programming but management science itself. The current multi-agent systems' "management vacuum" is the root cause of performance bottlenecks.
2. Constructed a systematic mapping framework from management theory to engineering implementation: We mapped Fayol's 14 principles to agent formation basic axioms, Weber's bureaucracy and Mintzberg's organizational configurations to agent topology and role design, Hersey-Blanchard's situational leadership to dynamic control logic for agent coordinators, and Drucker's MBO and Vroom's expectancy theory to objective functions and attention allocation mechanisms.
3. Proposed a bidirectional empowerment paradigm: AOM not only uses management to govern AI but uses AI to develop management. Large-scale agent simulation experiments will, for the first time, enable management science to become a computational science capable of controlled, reproducible, large-scale experiments.
4. Candidly defined the discipline's applicable boundaries: Identified inapplicable management theories (hierarchy of needs, two-factor theory, organizational politics, etc.) and proposed the fundamental design philosophy tension between control and emergence.
5. Depicted the ten-year vision for discipline and product: As a discipline, AOM has a clear research agenda and methodology; as a product, it will 催生 the Management-Architecture-Based Agent Workbench, compiling management philosophy into agent 底层 collaboration protocols.

We are witnessing a historic moment: management science expands from the study of human organizations to the engineering science of designing agent societies. Just as the 20th century's management revolution improved human organizational efficiency by orders of magnitude, the 21st century's Agent Organizational Management revolution will achieve equivalent leaps in agent collaboration efficiency.

This paper is the first map of this new continent. It is 注定 imperfect—as any foundational literature is. But it sets the direction, defines the coordinates, and establishes the grammar. We invite researchers from management science, computer science, organizational behavior, and complexity science to join this journey, collectively drawing more detailed maps.

The next century of management belongs to algorithms.

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Acknowledgments and Declarations

This paper is a Foundational Preprint, aiming to delineate research boundaries, establish theoretical frameworks, and 描绘 development visions for the emerging field of "Agent Organizational Management." The core viewpoints and frameworks of this paper are 首次 published here, and we welcome academic citation and discussion.

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We sincerely invite researchers and engineers from management science, computer science, organizational behavior, and complexity science to join this emerging field, collectively 推动 Agent Organizational Management from vision to reality.

This paper was written by Haoran Jiang, integrating perspectives from management science, artificial intelligence, organizational behavior, and complexity science. I believe the management science revolution of the AI era has only just begun.